

Remodeling of the Preoptic Area Drives Distinct Phases of Protandrous Sex Change

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Abstract

Phenotypic plasticity allows organisms to modify behavior, physiology, and morphology in response to changing environmental conditions. Among the most dramatic examples is post-maturational sex change, in which social cues trigger transformation from one functional sex to another during adulthood. In the protandrous anemonefish (*Amphiprion ocellaris*), males become females after attaining dominant social status, but the cellular mechanisms through which the brain orchestrates this transition remain poorly understood. Here, we combined single-nucleus multiomics (RNA and ATAC) with Visium HD spatial transcriptomics to characterize remodeling of the preoptic area (POA), a region which innervates the pituitary and regulates reproductive physiology, across multiple phases of sex change, including transitional individuals before and after gonadal transformation. Analysis of 35,364 nuclei revealed extensive cell-type-specific remodeling, with radial glia and POA neurons exhibiting the largest changes in abundance, gene expression, and chromatin accessibility. Machine-learning models revealed that sex change proceeds asynchronously across cell types and that many female-specific transcriptional programs emerge only after gonadal transformation. Radial glia displayed reduced expression of brain aromatase and transcription factor expression and chromatin accessibility patterns consistent with enhanced neurogenic activity during the intermediate phase. Concurrently, POA neurons showed evidence for increased incorporation of immature neurons, including expansion of a cholecystokinin-expressing population implicated in ovarian development. Together, these findings support a two-phase model in which early radial glial reprogramming and neurogenesis establish the neural substrate for sex change, while widespread feminization of pituitary regulating circuits follows ovarian development.

Introduction

The ability of the adult brain to reorganize itself in response to changing environmental conditions is one of the most remarkable adaptations in biology. Across vertebrate taxa, neuroplasticity allows animals to modify behavior, physiology, and reproduction in ways that improve fitness in an unpredictable world. Most examples, however, involve relatively modest adjustments in function, such as changes in learning, memory, motivation, stress responsiveness, or reproductive state.(CITATION NEEDED) Far less is known about how the adult brain coordinates large-scale functional reorganization of the entire organism. Understanding how the brain initiates and orchestrate such transformations remains a major challenge in biology.

Sequential hermaphroditism represents one of the most dramatic examples of organismal plasticity. In many coral reef fishes, social information perceived by the brain triggers a complete transformation from one reproductive sex to the other during adulthood. This transition requires coordinated remodeling of reproductive physiology, morphology, endocrine milieu, and behavior. As a result, sex-changing fishes provide a unique model to understand how the adult brain can direct extensive biological remodeling across multiple levels of organization, from gene expression and chromatin structure to physiology and behavior.

Most mechanistic studies of sex change have focused on protogynous species in which females transform into males following changes in social status. In these species, behavioral masculinization often occurs rapidly, within hours or days of social ascent, and is followed by progressive gonadal transformation and increased androgen production. Current models emphasize changes in neuroendocrine signaling, including alterations in aromatase activity and hypothalamic regulation of pituitary gonadotropins, which ultimately drive ovarian degeneration and testicular development. Once the gonads change, gonadal steroid hormones further reinforce male physiology, morphology, and behavior (1).

Protandrous sex change in anemonefish (*Amphiprion* spp.) appears to follow a fundamentally different trajectory (2). Following social ascent, sex-changing individuals can remain in an intermediate state for months before ovarian development occurs. During this period, male reproductive behaviors, circulating steroid milieu, and portions of the male gonad are largely retained. Rather than undergoing progressive feminization, the gonads enter a prolonged state characterized by reduced testicular tissue and immature ovarian tissue. Because a similar gonadal phenotype is observed in the reproductive partner of the sex-changing fish, this state may reflect reproductive stasis associated with the cessation of spawning rather than gonadal sex change itself. Only later does the remaining testicular tissue disappear vitellogenesis occurs (3). These observations suggest that, unlike protogynous species, substantial aspects of sex change in anemonefish occur before overt gonadal transformation.

Despite the apparent stability of the gonads during this prolonged intermediate period, previous work suggests that substantial neural remodeling is already underway. The preoptic area (POA), a conserved reproductive center that regulates pituitary gonadotrophs and reproductive physiology across vertebrates, has emerged as a particularly important region during protandrous sex change. Earlier studies demonstrated increased cell proliferation and neurogenesis within the POA during sex change, with evidence for a pronounced increase around the time of gonadal transformation. These findings suggest that neural reorganization precedes and potentially

contributes to subsequent gonadal feminization (4). However, the identities of the newly generated cells, and the extent to which neurogenesis contributes to functional remodeling of reproductive centers remain unknown.

A major challenge in understanding protandrous sex change has been the lack of cell-level information about the neural populations involved. Recently, we generated the first single-nucleus transcriptomic atlas of the sex-changing anemonefish forebrain, including the telencephalon and POA, and identified extensive sex differences across neuronal and glial populations. That study revealed widespread sex differences in gene expression, cell-type abundance, and predicted cellular communication networks. Particularly striking differences were observed among radial glial cells (RGCs) and POA populations implicated in gonadal regulation. These findings established the male and female endpoints of neural sexual differentiation and identified the POA as a major site of cellular remodeling during adult sex change (5).

However, the previous study compared only reproductively active males and females and therefore could not determine when these cellular differences emerged or whether they preceded or followed gonadal transformation. Consequently, the temporal sequence of neural remodeling during sex change remains unresolved. In particular, it is unknown whether the extensive changes observed in the POA arise during the prolonged period of reproductive stasis, during gonadal transformation itself, or only after female reproductive physiology has been established. Furthermore, because that analysis sampled the entire telencephalon together with the POA, only a limited number of nuclei from the POA were sampled. Although this population exhibited some of the most striking sex differences in the dataset, its limited representation constrained our ability to resolve changes in this key population. We therefore focused the present study on a targeted dissection centered on the POA and adjacent regions while removing most of the telencephalon, enriching representation of the POA in our dataset.

We combined single-nucleus multiomics with spatial transcriptomics to characterize the POA across multiple stages of protandrous sex change. By integrating transcriptional state, chromatin accessibility, spatial localization, endocrine phenotype, gonadal morphology, and behavior, we show the cellular and molecular progression of sex change from reproductive males through intermediate stages to reproductive females. Our analyses reveal stage-specific remodeling of RGCs and gonadotroph regulating cell populations that define the progression of sex change before and after overt gonadal transformation. Together, these findings provide a cellular framework for understanding how neural remodeling precedes and potentially orchestrates the transition to the female reproductive state.

Results

Male, Intermediate and Female Anemonefish Display Distinct Physiology and Behavior

To understand the process of protandrous sex change, we compared 6 reproductively active anemonefish pairs ($n = 6$ males and 6 females) to 10 pairs ($n = 20$) of male anemonefish housed together for 6 months (Fig. 1A, see methods). We paired males such that one male was notably larger than the other and was expected to occupy the α position in the social dominance hierarchy. These males (hereafter referred to as “intermediates”) are destined to undergo protandrous sex change, and given enough time, will become reproductively active females. The

partners of intermediates, males expected to remain males, are referred to as “intermediate partners”. We chose the 6-month timepoint as previous work in our lab has shown that at six months, approximately 10% of fish will have completed sex change, while the remaining fish will be in various intermediate stages determined by individual variation (3). Indeed, at this timepoint we found that one individual had completed sex change as indicated by the presence of vitellogenic oocytes (thus this individual was classified as a “New Female” and its partner as “New Male”) and two other individuals had a distinct gonadal phenotype defined by a large volume of empty space and non-vitellogenic oocytes. These individuals were classified as “Late Intermediates” and their partners as “Late Intermediate Partners”. Throughout this manuscript, we refer to these various stages of sex change as “phases”.

To assess the behavioral sex of the fish, we used a parental-care assay which has been shown to produce robust differences between males and females (3, 6–10). As expected, males performed significantly more parental care behaviors and spent significantly more time in the nest than their female partners (Fig. 1B). In agreement with previous results (3), intermediates retained male levels of parental care and prevented their partners from performing parental care. Late intermediates and their partners’ behavior aligned with female and male patterns of behavior respectively as did the new female and new male.

We next assessed the physiological sex of the fish. Intermediates (and Late Intermediates) displayed weights and lengths in between those of males and females indicating they were in the process of sex change (Figs. S1A, B). We next measured circulating levels of 11-ketotestosterone (11KT), the primary bioactive androgen in teleosts which displays extensive sexual dimorphism. Like behavior, intermediates retained male levels, while late intermediates and new females aligned with female levels. Late intermediate partners and new males aligned with male levels (Fig. 1B).

We next euthanized the fish and collected gonads for histological analysis. In contrast to the behavioral and hormonal measures, intermediates differed significantly from males in gonadal morphology. Specifically, intermediates exhibited a reduced proportion of testicular tissue, with a corresponding increase in proportion of immature ovarian tissue in their ovotestes. Further, intermediates had significantly lower total gonadal volume than males, suggesting testicular tissue was being lost while ovarian tissue remained (Fig. 1C). Importantly, this loss of testicular tissue is unlikely to reflect the onset of gonadal sex change, as intermediates and their partners remained similar in gonadal composition. Instead, it most likely represents gonadal regression associated with the temporary reproductive stasis that accompanies sex change (3). By the late intermediate stage, no measurable testicular tissue remained: the gonad consisted only of immature ovarian tissue, connective tissue, and empty space (Fig. 1D).

Multomics Capture High Quality Transcriptomes, Epigenomes, and Spatial Transcriptomes

Following euthanasia, we dissected out the ventrocaudal telencephala (Methods) targeting the POAs from males ($n = 6$), intermediates ($n = 7$), late intermediates ($n = 2$), the new female ($n = 1$), and females ($n = 6$) and used single-nucleus multomics (snMultiome; RNA + ATAC modalities) to profile gene expression and chromatin accessibility changes during sex change (see Fig. Sxx for details on how the dissection was performed). In total, 35,364 nuclei passed quality control (xxsee Methods) and were clustered using weighted shared nearest neighbors analysis. Hierarchical clustering yielded 27 clusters: 20 neuronal and seven glial (Fig. 2A).

Following clustering, we identified 10 glutamatergic (*slc17a6b+* or *slc17a7a+*), six GABAergic (*gad1+* or *gad2+*), and four mixed (GABA and glutamatergic markers) neuronal clusters. Further, we detected clusters of radial glial cells (1_RGC, *gfap+* and *crocc2-*), ependymal cells (15_EC, *gfap+* and *crocc2+*), dividing glia (26_DG, *gfap+* and *top2a+*), oligodendrocytes (2_OL, *mbpa+*), oligodendrocyte precursor cells (13_OPC, *cspg+*), microglia (11_MG, *ptprc+* and *p2ry12+*) and leukocytes (20_Leuko, *ptprc+* and *p2ry12-*). These markers along with strong markers for each cluster from Seurat's "FindMarkers" function are shown in Figure 2B.

Using Visium HD, we further generated spatial transcriptomic atlases of anemonefish telencephala and adjacent regions using n = 6 male and 6 female anemonefish. In total 433,999 cell segmented polygons passed quality control. We then manually annotated neuroanatomical regions (methods, Fig. S2). To annotate our snMultiome clusters, we next used Cell2location to project the multiome clusters onto our spatial atlases (11) (Fig. 2C,D). Figure 2C shows a heatmap of the cell type composition of each region. We labeled clusters based on primary cell type (neuronal versus glial), primary neurotransmitter (Glutamate, GABA, or mixed), and neuroanatomical location if the cell type was spatially localized (see Appendix SX for all cell type labels and SX for abbreviations).

Several multiome clusters mapped to specific neuroanatomical regions. Cluster 6 mapped strongly to parvocellular and magnocellular regions of the POA. This mapping along with expression of *hmx2*, *hmx3a*, *hmx3b*, *avp*, *oxt*, *cckb*, *otpa* (12–15) lead us to annotate this cluster as the POA (6_POA_Mixed). Clusters 0, 3, 14, 21, and 24 mapped to other subpallial regions of the telencephalon (0_Spall_Mixed, 3_Spall_GABA, 14_Vsst_GABA (16), 21_Spall_Gaba, 24_Vs-M_Mixed) while Cluster 7 mapped to pallial regions (7_Pall_Glut). Clusters 4, 8, 10, 19, 22, and 23 mapped to diencephalic regions (4_TGN_Glut, 8_VOT_Glut, 10_nH_Glut, 19_Di_Glut, 22_Di_Glut, 23_NLT_GABA). Drawings of VisiumHD sections with neuroanatomical regions annotated are shown in Fig. S2).

To further characterize our cell-types, we cross-projected cluster labels from this snMultiome dataset with our previously published single-nucleus RNA-sequencing (snRNA-seq) atlas of the anemonefish forebrain (Fig. SXX). We also used this opportunity to spatially annotate the clusters from our snRNA-seq atlas (Fig. SXX). Cell populations showed strong correspondence between datasets despite differences in tissue sampling and sequencing modalities (Fig. SXX). In particular, the POA cluster 6_POA_Mixed was the mutual best match (or something) for cluster 13d described in Parker et al., 2024, and cluster 13d independently mapped to the POA in our Visium HD data (Fig SXX) These findings demonstrate the robustness and reproducibility of the identified cell types while establishing continuity between the present atlas and our previous work.

Asynchronous Changes in Cell Type Composition During Sex Change

Because sex change is associated with changes in brain size and composition (4), we next analyzed changes in cell populations across sex change. We detected significant phase differences in the abundance of cells in 12 of the 27 multiome clusters. Of these, nine were relative decreases in cell population while only three were increases. Notably, all three increases occurred in glia. Oligodendrocytes (2_OL) increased in population progressively with each phase of sex change, and microglia (11_MG) increased in population early in sex change during the male-intermediate transition (Fig. S3A). Consistent with the idea that sex change is characterized by transient increases in neuro- and gliogenesis (4, 5), the proportion of radial glia (1_RGC)

transiently increased in intermediates before returning to baseline levels in females (Fig 3A). Together, these findings reveal asynchronous changes in glial populations during sex change, with an early expansion of neurogenic RGCs and microglia followed by later accumulation of oligodendrocytes. This temporal sequence is consistent with RGCs supporting the production of new neurons early in sex change and microglia supporting and maintaining the expanded population, followed by oligodendrocytes myelinating the new neurons as they integrate into circuits.

Because the large-scale changes in glial populations described above may obscure more subtle changes in neuronal populations, we next analyzed changes in proportion only considering neurons. Of the 20 neuronal clusters we analyzed, 11 changed in population, with four decreasing and seven increasing in population over the course of sex change. During the male-intermediate transition, we found significant increases in clusters 14_Vsst, 17_Glut, and 18_Glut. In intermediates, we found a transient increase in cluster 0_Spall_Mixed, and a transient decrease in cluster 4_TGN_Glut. Interestingly, we found a significant increase during the intermediate-female transition in clusters 6_POA_Mixed (Fig. 3B) and 19_Di_GABA and decreases in 9_Mixed, 22_Di_Glut, and 24_Vs-m_Mixed (Fig. S3B). In contrast, other clusters underwent early transitions, exhibiting significant increases between males and intermediates but no further change between intermediates and females. These early increases were observed in clusters 14_Vsst, 17_Glut, and 18_Glut (Fig. S3B). We also identified transient changes that were specific to the intermediate stage, including an increase in cluster 0_Spall_Mixed and decreases in clusters 4_TGN_Glut, 19_Di_GABA, and 7_Pall_Glut.

Together, these findings indicate that neuronal remodeling during sex change is a staged process. Many neuronal populations showed early or transient changes in proportion before the completion of gonadal transformation, suggesting that extensive neural remodeling begins well in advance of the endocrine and reproductive transition. By contrast, the delayed expansion of the POA cluster 6_POA_Mixed, a cell-type that plays a central role in regulating the hypothalamic-pituitary-gonadal axis, was not evident until after gonadal transformation and the accompanying changes in circulating sex hormones. These findings suggest that distinct neuronal populations remodel at different stages of sex change, with some populations potentially contributing to the initiation of the transition, whereas others remodel only after the new reproductive state has been established.

Cell Types Undergo Sex Change at Different Rates

Transcriptomic sex change

To characterize the transcriptomic changes associated with sex change, we next identified differentially expressed genes (DEGs) and classified each according to its temporal pattern and direction of expression changes across males, intermediates, and females (Methods). Specifically, genes were classified as exhibiting early, late, progressive, or transient patterns as well as either increasing or decreasing expression. We identified a total of 290 DEGs (Appendix SX) DEGs were roughly split, with 150 DEGs being upregulated and 140 being downregulated going from male to female. Temporally, there were 76 DEGs showed an early pattern of change, 136 late changes, 6 progressive and 72 transient. The predominance of late changing DEGs indicates that the greatest amount of transcriptomic remodeling occurs during the later stages of sex change following gonadal transformation. We found DEGs were biased in clusters 6_POA_Mixed (Fig. 3C), 1_RGC, 0_Spall_Mixed, and 11_MG.

Epigenomic sex change

To characterize the epigenomic changes associated with sex change, we next identified differentially accessible regions (DARs) and classified each according to the same temporal patterns described above. We identified a total of 53 DARs (Appendix SX), of which 6 exhibited early changes, 22 late changes, 25 progressive changes, and none exhibited transient changes. Overall, 28 DARs decreased in accessibility and 25 increased in accessibility. Similar to the transcriptomic analysis, the predominance of late DARs indicates that the greatest amount of epigenomic remodeling occurs during the later stages of sex change. Unlike gene expression, however, chromatin accessibility was characterized by progressive changes and an absence of transient remodeling, suggesting a more gradual trajectory of epigenomic reorganization. We next asked whether specific cell populations underwent greater epigenomic remodeling than expected by chance by testing whether clusters contained more DARs than expected based on the total number of accessible regions analyzed within each cluster. This analysis identified significant enrichment of DARs in 1_RGC, 0_Spall_Mixed, and 2_OL (Fig. 3D), indicating that these cell populations undergo disproportionately extensive epigenomic remodeling during sex change.

Relationship Between DEGs and DARs

We next asked which changes in gene expression could be explained by changes in the epigenome, or whether epigenomic changes could explain changes in gene expression. A direct comparison of differential gene expression and differential chromatin accessibility identified only a single overlapping gene (*igf2bp1* in cluster 0_Spall_Mixed). Because this comparison requires both transcriptional and accessibility changes to independently satisfy strict significance thresholds, we next asked which DEGs and DARs had a significant correlation between the alternate modality (i.e., whether the expression of a DEG is correlated with its chromatin accessibility and vice versa). To do this, we correlated pseudobulked gene expression with pseudobulked chromatin accessibility scores for ATAC peaks located within promoter and gene-body regions of each gene across individuals (Methods). Following quality control (Methods) 226 of the 290 DEGs were suitable for correlation analysis (Appendix SXX, SXX).

Of the 226 genes we analyzed, 67 exhibited a significant correlation between gene expression and the accessibility of their best local ATAC peak (Pearson's $P < 0.05$). The prevalence of accessibility-associated DEGs did not differ significantly among cell clusters (likelihood-ratio test, $\chi^2_{19} = 15.57$, $P = 0.686$), indicating that the contribution of local chromatin accessibility to transcriptional remodeling was broadly similar across cell populations.

As a complementary analysis, we next examined whether DARs were associated with expression of their nearest expressed gene. Following quality-control 23 of the 53 DARs were suitable for analysis. Of these, four exhibited significant correlations between chromatin accessibility and gene expression (Pearson's $P < 0.05$) (Appendix SXX, SXX).

Together, these analyses indicate that although relatively few genes simultaneously met stringent criteria for both differential expression and differential accessibility during sex change, local chromatin accessibility markedly predicted transcriptional remodeling (DEGs were predicted by chromatin), while transcriptional remodeling did not predict chromatin accessibility (DARs were not predicted by transcription). These findings support a substantial contribution of chromatin

accessibility to transcriptional remodeling while also indicating that additional regulatory mechanisms influence the expression of many genes.

Sex Change Occurs Asynchronously Across Cell Types

To determine a temporal order for when sex change occurred in each cluster, we next trained machine learning models to predict the sex of each cluster for each intermediate based on the expression of DEGs and accessibility of DARs that differed between males and females (methods). Because intermediates were all sampled at the same time, the mean prediction of the intermediates' sex in a given cluster reflects the temporal order of when the cluster changes during sex change. A prediction of male (closer to 0) indicates the cluster has not changed sex by six months, while a prediction of female (closer to 1) indicates it has. Of the 27 clusters, 16 passed quality control metrics and were analyzed (methods). The mean score of analyzed clusters was 0.340 ± 0.063 SE (Appendix SX) implying that sex change is approximately 1/3 complete at six months. Broadly, GABA and glutamatergic neurons tended to change sex before mixed neurons and glia. The final clusters to change sex in our analysis (i.e. the most male-like in intermediates) were 1_RGC, 6_POA_Mixed, 3_Spall_GABA (Fig. 3E). Taken together, these data support the idea that sex change occurs asynchronously across cell types. Further, they suggest that the clusters that undergo the greatest change during sex change (1_RGC, 6_POA_Mixed, 0_Spall_Mixed, and 11_MG) remain in a largely male-like state until after gonadal sex change occurs.

The Intermediate State is Characterized by a Decrease in Estrogenic Signaling from Radial Glia

Among all identified cell populations, cluster 1_RGC exhibited the second-highest number of DEGs and the greatest number of DARs. Further, sex change is associated with extensive changes in cell population and steroid hormone signaling in the brain (2, 5, 17–22). Since radial glial cells (RGCs) are the neural progenitors responsible for neurogenesis, as well as the primary cell type responsible for steroid hormone synthesis in the teleost brain (23), we next focused our analysis on cluster 1_RGC.

Among the most prominent changes in 1_RGC was transient downregulation of brain aromatase (*cyp19a1b*), the enzyme responsible for converting androgens to estrogens in the teleost brain (23). *cyp19a1b* expression was significantly reduced in intermediates relative to both males and females, whereas males and females did not differ (Fig. 4A). This is in contrast to previous findings which have shown that across sex change, aromatase increases gradually along the process (17), suggesting that a reduction in E2 production may be specific to the ventrocaudal telencephala, while E2 production increases linearly when the whole brain is analyzed.

Evidence for Increased Neurogenesis During the Intermediate Phase

E2 signaling is known to inhibit stem cell differentiation and neurogenesis in zebrafish, (24, 25) and in mammals is known to promote brain masculinization during development (26, 27). In our DEG analysis, the largest single group of DEGs in 1_RGC were transcription factors known to play roles in neuro- and gliogenesis or genes related to the epigenome (28–38) (Fig.

4B). Additionally, our DAR analysis in 1_RGC detected differences in several transcription factors and cell adhesion proteins (Fig. S4). Given these observations along with the downregulation of *cyp19a1b* and changes in neuronal cell populations, we next examined how neurogenesis changed over the course of sex change. Because RGCs are a heterogeneous cell type, we subclustered 1_RGC into 4 subclusters (Fig. 4C). Analysis of marker genes paired with CytoTRACE, a tool that infers cellular immaturity(39) revealed a neural stem cell (NSC) population (1_NSC), a neural progenitor (NP) population (1_NP), as well as *shha* and *gfap* expressing populations (1_shha and 1_gfap respectively) (Figs. 4D,E, S5A-D).

There were significant changes in the populations of 1_NSC and 1_NP by phase. In intermediates, the population of 1_NSC was significantly depleted relative to males and females (Fig. 4F). Conversely, intermediates had the largest population of 1_NP, while females had the smallest (Fig. 4G). To assess the neurogenic state of 1_NP in each phase, we created a “neuron differentiation” gene expression module consisting of genes annotated with that GO term (GO: 0030182, methods). Males and intermediates expressed this module at similar levels, while females expressed this module significantly less (Fig. 4H). This arrangement supports a model where the NSC population is depleted in intermediates at a higher rate than it is replenished and these NSCs become NPs. In intermediates, this expanded NP population undergoes neurogenesis at a similar rate as it did in males leading to an overall increase in neurogenesis in the intermediate phase. During the intermediate-to-female transition, the rate of neurogenesis decreases as does the population of 1_NP.

Evidence for Neurogenesis and Neuronal Maturation in 6_POA_Mixed

Given the changes in cell population (Fig.3B), late sex change prediction (Fig.3E), and large number of DEGs (Fig. 3C), we next focused our analysis on 6_POA_Mixed (Fig. 5A). GO analysis of the DEGs in 6_POA_Mixed revealed several genes related to the perineuronal net change during sex change (Fig. S6). Further several genes associated with the extracellular matrix (ECM) and cell adhesion were found to be DARs or DEGs (Fig.5B, C)(40–47). Given the changes in neurogenic RGCs as well as previous data showing neurogenesis occurs in the anterior POA during sex change(4), we hypothesized that changes in neurogenesis populating 6_POA_Mixed were occurring and these DEGs were indicative of the integration of new cells into 6_POA_Mixed.

To test the hypothesis that DEGs related to the ECM or cell adhesion (ECM-DEGs) in 6_POA_Mixed were indicative of new cells integrating into local circuits, we compared the relative maturity of cells expressing ECM-DEGs to those not expressing ECM-DEGs (methods). Indeed, ECM-DEG+ cells were significantly less mature than their ECM-DEG- counterparts, and females had significantly more mature ECM-DEG+ cells than males and intermediates (Fig.5D). In 6_POA_Mixed as a whole, intermediates had significantly less mature cells than females suggesting that intermediates in general are characterized by lower cellular maturity than males or females (Fig. 5E). As further evidence for the hypothesis that new cells are added during the intermediate state, intermediates showed significantly higher expression for a module consisting of genes annotated with the GO terms “brain development” (GO:0007420) (Fig. 5F) and “axon guidance receptor activity” (GO:0097484) (Fig. 5G). Taken with the changes in cell population in 6_POA_Mixed, these findings support a model where new cells are added to 6_POA_Mixed during the male-intermediate transition, but do not yet mature until the intermediate-female transition. As they mature and integrate into their circuit, they express ECM-DEGs.

Candidate Gonadotroph-Regulating Neurons Undergo Distinct Modes of Remodeling During Sex Change

Having identified evidence that newly generated neurons are incorporated into 6_POA_Mixed, we next asked which molecularly defined neuroendocrine populations account for this remodeling. Specifically, we sought to distinguish neuronal populations that expand through incorporation of newly generated neurons from those that remodel primarily through changes in gene expression within preexisting neurons.

We detected significant changes in several molecularly defined neuronal populations implicated in gonadotroph regulation, identified by expression of *drd3*, *tacr3a*, *np7r*, *nmb*, *cckb*, or *gnrh1* (Fig. 5C; Fig. S7) (48–53) (54). During the intermediate-to-female transition, both the proportion of *drd3*-expressing neurons and *drd3* expression increased, while *tacr3a* expression and the proportion of *tacr3a*-expressing neurons increased transiently during the intermediate phase before declining below male levels in females (Fig. 6A,B). In contrast, *np7r* and *nmb* expression decreased during the male-to-intermediate transition without corresponding changes in the abundance of marker-positive neurons, whereas neither *gnrh1* expression nor the proportion of *gnrh1*-expressing neurons changed across sex change (Fig. S7). Unlike these patterns, the proportion of *cckb*-expressing neurons increased significantly during the intermediate-to-female transition despite no detectable change in *cckb* expression (Fig. 6C). This suggests POA *cckb* neurons might remodel through a mechanism distinct from the other candidate neuroendocrine populations.

For the cell types that increased during sex change, we further compared the predicted maturity of marker-positive cells across phases to determine whether these changes were more consistent with incorporation of newly generated neurons or transcriptional remodeling of preexisting neurons. Neither *drd3*- nor *tacr3a*-expressing neurons differed in maturity across sex change (Fig. 6A,B), suggesting that the increased abundance of these populations is more consistent with transcriptional reprogramming than incorporation of newly generated neurons. Similarly, cells expressing the steroid hormone receptors *pgr* and *ar*-like increased in abundance and expression during the intermediate-to-female transition but showed no change in predicted maturity (Fig. S7), further supporting transcriptional remodeling of preexisting neurons rather than incorporation of newly generated cells.

In contrast, *cckb*-expressing neurons were significantly less mature in males and intermediates than in females (Fig. 6C), consistent with progressive maturation of newly generated neurons after their incorporation into the preoptic circuit during the intermediate-to-female transition. Together with their expansion during the intermediate-to-female transition and progressive maturation across sex change, these findings identify *cckb*⁺ neurons as the strongest candidate population generated by neurogenesis in the POA during sex change.

Discussion

Here, we combined single-nucleus transcriptomics, chromatin accessibility profiling, and spatial transcriptomics to generate a cell-type-resolved molecular atlas of socially induced protandrous sex change in *Amphiprion ocellaris*. By integrating these datasets with behavioral observations, circulating steroid hormones, and gonadal histology, we reconstructed the temporal progression of sex change from male through intermediate stages to mature female. This approach revealed extensive remodeling of both neuronal and glial populations, identified radial glial cells and the preoptic area as the principal sites of molecular reorganization, and provides a

mechanistic framework linking socially induced changes in the brain to the endocrine events that ultimately drive gonadal sex change.

Among all cell populations, radial glia underwent the most extensive molecular remodeling during sex change, exhibiting the second-highest number of differentially expressed genes and the greatest number of differentially accessible chromatin regions. At first glance, these findings might suggest that radial glia rapidly acquire a female-like phenotype. Surprisingly, however, our machine-learning analyses indicated the opposite: after six months of social ascent, radial glia remained among the most male-like cell populations in the brain. This apparent paradox reflects the complex temporal organization of radial glial remodeling. Rather than transitioning directly from a male to a female transcriptional state, radial glia exhibited a mixture of early, transient, progressive, and late transcriptional programs together with predominantly progressive and late changes in chromatin accessibility. These observations suggest that socially induced sex change does not involve a simple linear feminization of radial glia. Instead, radial glia appear to enter a prolonged transitional state characterized by extensive transcriptional and epigenetic reprogramming while maintaining much of their original cellular identity. Given their established role as the principal neural stem cells of the teleost brain, this prolonged transitional state may provide the cellular substrate through which socially induced signals initiate the neurogenic and neuroendocrine remodeling that precedes gonadal transformation.

One of the earliest and most striking features of this transitional radial glial state was the transient reduction in expression of brain aromatase: *cyp19a1b*. Because radial glia are the principal source of aromatase in the teleost brain, this finding suggests that local estrogenic signaling within the ventrocaudal telencephalon is selectively reduced during the intermediate phase of sex change. In zebrafish, radial glia similarly downregulate aromatase as they enter regenerative neurogenic programs following brain injury (xx24), raising the possibility that transient suppression of estrogen synthesis represents a conserved feature of radial glial activation. Our observations therefore suggest that reduced aromatase expression may help establish the permissive cellular environment required for the increased recruitment of neural stem cells into the neurogenic lineage that we observed during the intermediate phase. The high expression of estrogen receptors within the 6_POA_Mixed population further suggests that these neurons are well positioned to respond to the local reduction in estrogen signaling associated with *cyp19a1b* downregulation. Although previous whole-brain transcriptomic analyses reported progressive increases in brain aromatase expression during protandrous sex change (xx16), the apparent discrepancy may reflect regional specialization that becomes obscured in bulk tissue analyses, differences in sampling stage, or species-specific differences between *A. ocellaris* and *A. bicinctus*. Among these possibilities, the regional specificity revealed by our cell-type-resolved analysis appears the most likely explanation.

The preoptic area emerged as the principal neuronal target of this radial glial remodeling. Previous work from our laboratory demonstrated that neurogenesis increases within the POA during protandrous sex change, but the identities and developmental progression of the newly generated neurons remained unknown. Our results provide multiple independent lines of evidence that these newborn neurons are incorporated into preoptic neuroendocrine circuits. The POA exhibited the greatest transcriptional remodeling of any neuronal population and underwent a progressive increase in cell abundance across sex change. In addition, extracellular matrix remodeling, increased expression of developmental and axon-guidance gene modules, and CytoTRACE analyses identified an expanded population of relatively immature neurons during

the intermediate phase that subsequently matured in females. Together, these findings suggest that POA remodeling occurs in two temporally distinct phases: an early phase characterized by neurogenesis and neuronal differentiation, followed by a later phase in which these neurons acquire mature transcriptional identities and become incorporated into the female neuroendocrine network. Although extracellular matrix remodeling could also reflect structural plasticity independent of neurogenesis, the convergence of changes in cell abundance, developmental gene expression, cellular maturity, and our previous lineage-tracing study strongly supports increased neurogenesis as a major contributor to POA remodeling during sex change.

In addition to radial glia and POA neurons, microglia also underwent disproportionately extensive transcriptional remodeling during sex change. Although the functional significance of these changes remains unclear, microglia are well established as regulators of adult neurogenesis, neuronal maturation, synaptic remodeling, and the elimination of neurons and synapses during circuit refinement. Given that several neuronal populations declined in abundance while others expanded during sex change, microglia may contribute both to the maturation of newly generated neurons and to the remodeling or elimination of neuronal populations that are no longer required as reproductive circuits transition to the female state.

Among the POA neuroendocrine populations undergoing remodeling, CCKB-expressing neurons emerged as the strongest candidate for the newly generated neuronal population. This cluster exhibited one of the largest increases in cell abundance during sex change, displayed an immature transcriptional profile during the intermediate phase, and acquired a mature female-like state only later, consistent with the developmental progression inferred from our CytoTRACE analyses. Recent studies in teleosts have identified cholecystokinin signaling as a major regulator of pituitary FSH secretion, functioning in many respects as a gonadotropin-releasing system specifically for the FSH axis. Because FSH is essential for ovarian growth, follicular development, and vitellogenesis, the selective expansion and maturation of CCKB-expressing neurons during sex change provides a plausible mechanistic link between socially induced neurogenesis and activation of the female reproductive axis. Although additional functional studies will be required to establish causality, our findings raise the possibility that radial glia-generated neurons are preferentially incorporated into the CCKB neuroendocrine pathway, thereby coupling social status to ovarian development.

Additional changes within the POA further support coordinated remodeling of the pituitary gonadotroph axis during sex change. In addition to the expansion of CCKB-expressing neurons, multiple neuroendocrine pathways converged on a pattern consistent with increased FSH signaling and reduced LH signaling as fish transitioned to the female state. Receptors for the LH-stimulatory neuropeptides neuropeptide Y (NPY) and neurokinin B (NKB) were downregulated, whereas expression of the dopamine receptor, a pathway associated with inhibition of LH release in teleosts, increased. Together, these coordinated changes suggest that remodeling of the POA is not limited to the generation of new neuroendocrine neurons but involves a broader reorganization of signaling pathways that collectively shift pituitary regulation from a male- toward a female-typical endocrine state.

Collectively, our findings support a model in which socially induced sex change proceeds through two temporally distinct phases of brain remodeling (Fig. XX). During the intermediate phase, social ascent rapidly activates radial glia, resulting in transient downregulation of *cyp19a1b*, recruitment of neural stem cells, and the production of immature neurons that populate the preoptic area. At this stage, many of the molecular programs associated with neurogenesis, neuronal differentiation, and circuit assembly are initiated, yet the newly generated

neurons remain relatively immature and the POA retains much of its male-like molecular identity. During the subsequent transition to the female state, these neurons acquire mature transcriptional profiles while neuroendocrine pathways are remodeled in a manner consistent with increased FSH signaling and reduced LH signaling, coincident with ovarian differentiation and changes in circulating sex steroids. This temporal sequence raises the possibility that the intermediate stage represents a developmental program that establishes the cellular substrate for female neuroendocrine circuits, whereas subsequent endocrine feedback from the transforming gonad contributes to their maturation and stabilization. Rather than representing a simple linear feminization of the brain, our findings suggest that socially induced sex change is orchestrated through sequential interactions among radial glia, neurogenesis, neuroendocrine remodeling, and the gonad.

Several important questions remain. Most notably, the identity of the neurobiological signal that initiates radial glial remodeling is unknown, as is the mechanism by which changes in gonadal steroid production influence the maturation of newly generated neurons. We further predict that the newborn neurons populating the POA are enriched for CCKB-expressing neuroendocrine cells, a hypothesis that can now be tested directly using lineage tracing and functional manipulation. Resolving these questions will provide a mechanistic understanding of how social information is transformed into stable changes in reproductive neuroendocrine circuitry.

More broadly, our findings suggest that socially induced sex change is not accomplished through the direct conversion of existing neural circuits but through a temporally ordered program of developmental remodeling. Rather than functioning solely as support cells, radial glia emerge as central integrators of social information, initiating a cascade of neurogenesis and neuroendocrine reorganization that ultimately reshapes reproductive physiology. This framework raises the possibility that prolonged intermediate states, observed in many sequentially hermaphroditic teleosts, represent active developmental phases during which new neural circuits are assembled before they are functionally incorporated into the mature reproductive network. Understanding how social cues initiate this developmental program and how gonadal feedback subsequently stabilizes these newly formed circuits will provide important insights into the mechanisms by which the adult vertebrate brain reorganizes in response to changing social environments.

Methods

Animals and Husbandry

Anemonefish (*Amphiprion ocellaris*) used in this study were first- or second-generation anemonefish obtained from Oceans, Reefs, and Aquariums (Fort Pierce, FL). All fish were housed in either 20-gallon tall (24" x 12" x 16") or 25-gallon cube (18" x 18" x 18") centrally filtered aquariums with a 6" terra cotta pot serving as a nest site and spawning substrate. In all tanks, salinity was kept at 1.026 specific gravity, temperature was kept between 26°-28°C, pH was kept between 8.0 and 8.4, and lighting was provided by fluorescent and LED lights over tanks on a 12:12 light schedule (lights on at 07:00 h and off at 19:00 h). Fish were fed twice daily with ReefNutrition TDO Chroma Boost pellets (Reef Mariculture Inc., Campbell, CA).

Initiation of Sex Change

We induced sex change by creating eight pairs of anemonefish (n = 16) comprised of two males that had previously spawned with females. We paired males such that one male was

notably larger than the other and would thus occupy the α position in the dominance hierarchy. We further induced sex change in two additional pairs ($n = 4$) by removing the α female from a triad (a group of fish consisting of 1 α female, one β male, and 1 γ male) such that the β male would ascend to the α position of the dominance hierarchy and the γ male would occupy the β position. In total, this resulted in the creation of 10 sex changing pairs ($n = 20$). Here, the larger, more dominant male (hereafter referred to as a “intermediate”) is destined to undergo protandrous sex change and become a female, while its partner (hereafter referred to as a “intermediate partner”) is destined to remain a male.

On the day of pairing or female removal, intermediates and subordinates were weighed and measured. Following pairing, pairs were given six months to undergo sex change. We chose six months as previous work in our lab has shown that at the six-month time point, approximately 10% of intermediates will have completed sex change and become females (as indicated by the presence of vitellogenic oocytes), while the remainder will be in various intermediate stages of sex change(3).

Phases

After six months, there is significant variation in the progression of sex change in intermediates. We discovered that one intermediate had fully completed sex change, and two intermediates had a distinctly expanded gonad with no testicular tissue and non-vitellogenic oocytes. We reclassified these intermediates as “New Female” and “Late Intermediate” respectively. Similarly, we reclassified their partners as “New Male” and “Late Intermediate Partner” respectively. In all statistical analyses, these individuals are analyzed separately to the other intermediates and intermediate partners. Throughout this manuscript, we refer to these classifications of sex change as “phases”. In all experiments, these individuals were compared to the same 6 pairs of males and females ($n=12$) which served as controls.

Computation

All computational analyses described below were performed in R 4.4.2 or Python 3.12.4. In all cases unless otherwise stated, we considered a p -value of <0.05 (or false discovery rate q -value (55) where relevant) to be statistically significant. For pairwise comparisons, we used the “emmeans” package with no p -value adjustment specified (56). Type-III tests for mixed effects (lmer) models were computed using the “Anova” function from the package “car”(57), and for fixed effects models (lm and glm) were computed using the R’s “anova” with test = “Chisq” specified .

Parental Care Assay

Six months following the initiation of sex change, we assessed male-like behaviors using a parental care assay known to produce robust and reproducible sex differences in anemonefish as described previously (3, 6–10). To do this, we replaced the terra cotta pot serving as the pair’s nest with a terra cotta pot containing eggs from another pair. Anemonefish do not discriminate between their own eggs and unrelated eggs, and parent them with similar vigor to their own(6, 10). We then allowed 24 hours for the fish to acclimate to the new pot. One hour prior to recording, we placed a camera in front of the tank to allow acclimation. We then recorded parental care behaviors (fanning, nipping, and wiggling) as well as the time spent in the nest for

10 minutes (Appendix SX). We analyzed these data using two linear regressions, one fitting the time spent in the nest to phase, and another fitting the sum of parental care behaviors to phase.

Serum Hormone Analysis

Following behavioral analysis, blood was drawn from individuals' caudal veins using a 0.9% heparinized needle. Samples were stored on ice and then centrifuged at 1250 RCF at 4°C for 15 minutes. Plasma was stored at -80°C prior to analysis. We then assessed 11KT levels using a Cayman Chemical 11KT kit (Cayman Chemical, #582751). Plasma samples were diluted to 1:30 in assay buffer and analyzed following manufacturers guidelines using an Epoch Microplate Spectrophotometer (BioTek Instruments) at 412 nm. We analyzed all samples as duplicates.

We were unable to measure the plasma 11KT from two individuals, one male and one female; these individuals were omitted in analyses of 11KT levels. Phase differences in 11KT levels were analyzed using a linear regression fitting log10 transformed 11KT values by phase.

Gonads

Individuals were then weighed and measured again. Individuals were then euthanized via decapitation, and the body was stored in 4% paraformaldehyde (PFA) at 2°C for 48 hours. Following PFA fixation, bodies were cryoprotected in 30% sucrose for 48 hours at 2°C. Bodies were then mounted in OCT (Tissue-Tek, Sakura Finetek) and stored at -80 °C. Gonads were then cryosectioned in the sagittal plane at 40µm (Microm HM 550, Thermo Scientific), mounted on charged slides, and stained with hematoxylin and eosin using standard procedures.

To analyze the gonadal composition of individuals (excluding females and the new female which only have ovarian tissue(58)), 20 random sections from each individual were imaged using a Zeiss Axioskop 40 (Carl Zeiss Microscopy LLC, White Plains, NY) at 10x magnification. These images were analyzed in Photoshop by selecting regions of testicular tissue, ovarian tissue, connective tissue, and recording the number of pixels of each type. We then calculated the proportion of gonad that was testicular or ovarian by dividing the total respective area by the total recorded gonad area (gonadal + ovarian + connective) excluding empty space. Proportion of testicular or gonadal tissue was analyzed using a linear regression fitting the proportion of gonad made up of the associated tissue type by phase.

To quantify gonadal volume, we also imaged a random 20 sections of gonad at 2.5x magnification using a Zeiss AxioScope A1 (Carl Zeiss Microscopy LLC, White Plains, NY) such that the entire gonad was within frame. If this was not the case, several images were taken. We then used Photoshop to measure the area of the gonad in each slide. From this sample, we were able to estimate the total volume of the entire gonad in pixels for each individual. Due to large differences between the phases, we log10 transformed this value in all analyses. Because, gonadal volume is correlated with body size, we analyzed gonadal volume with phase and body length as covariates. For the bottom panel in Figure 1B, we plot the residuals of a linear regression only including body length as a fixed effect such that the phases could be directly comparable.

POA Dissection

Following decapitation, the brain was exposed and immediately treated with chilled dissection medium by pipetting a small volume over the dorsal surface. Dissection medium consisted of Hibernate AB complete medium (with 2% B27 and 0.5 mM Glutamax; BrainBits, LLC, Springfield, IL) with 200 U/mL of Protector RNase Inhibitor (Roche, Basel, Switzerland). The brain was then removed and placed in a petri dish containing chilled dissection medium. We then isolated the ventrocaudal telencephala (see supplementary ... for details) and placed them in RNase free tubes and stored them at -80°C.

snMultiome

...

Spatial Transcriptomics

...

Neuroanatomical Annotation

Because there is no published atlas for any anemonefish brain, we relied on a published atlas from *Astatotilapia burtoni* (59) in combination with unbiased clustering and gene expression (see “Spatial Transcriptomics”) for our annotations. Interestingly, we noted a few differences between the *A. ocellaris* and *A. burtoni* brains. Transcriptomically, Dl-v2 (ventral part of the lateral zone of the dorsal telencephalon, subdivision 2) and Dl-g (granular zone of lateral zone of the dorsal telencephalon), were distinct in their transcriptomes depending on rostral versus caudal position or dorsal versus ventral position. Thus, we reannotated them as Dl-v2d and Dl-gd (dorsal part). We also noted a neuronal cluster of high cell density in the dorsolateral pallium near Dl-g that appeared to have no equivalent in *A. burtoni* and had unique rostral and caudal transcriptomes. Thus, we labeled this dorsolateral nucleus rostral and caudal parts (Dln-r and Dln-c) (Figure Sxx). Because this cell type was largely dissected out in our dataset, we did not examine it farther.

Projection of snMultiome Data Onto VisiumHD Data

To map our snMultiome clusters onto the VisiumHD sections, we used Cell2location (11). To do this, we first trained regression models (RegressionModel) on raw RNA counts excluding genes that were present in fewer than 10 cells. The model was conditioned on sample/pool identity as a batch covariate and on cluster identity as the cell type label. The model was trained for 250 epochs with batch size set to 4096 and posterior distributions were exported using 1000 posterior samples. Mean expression per cell cluster was extracted from the posterior (means_per_cluster_mu_fg) to construct the reference signature matrix for downstream mapping. Next, we fit the model to the VisiumHD sketch assay (bin level) only including bins that would have been included in our dissection of the ventrocaudal telencephalon during snMultiome sequencing. Genes were intersected between the spatial sketch dataset and the reference signature matrix to restrict analysis to a shared gene set. The cell2location spatial mapping model (Cell2location) was then fit to the sketch data, using slice identity as a batch covariate, an expected cell abundance per spot prior of $N_{cells_per_location} = 3$, and a detection_alpha of 200 to account for technical variation in detection sensitivity across slices. The model was trained for up to 1,000 epochs (batch size 2048, full training set, no held-out validation split). Posterior cell type abundances were exported using 100 posterior samples, yielding per-spot mean, 5th percentile (q05), and 95th percentile (q95) estimates of absolute cell type abundance for each reference-derived cell type. For each sketch spot, the cell type with the highest 5th-percentile (q05) posterior abundance estimate was assigned as the spot's dominant cell type. These categorical dominant-cell-type labels were then projected from the sketch assay to the full-resolution Visium HD dataset using Seurat's ProjectData function, which maps sketch-level

metadata onto the full dataset via a shared PCA embedding (15 dimensions). This yielded a full-resolution spatial map of predicted dominant cell type identity across all tissue bins.

Cell Type Annotation

We annotated our snMultiome clusters using a combination of *a priori* marker genes (see Results) and spatial mapping from Cell2location. We annotated cell types first by cluster number, then by primary cell type, (e.g., 1_RGC = Radial Glial cell, 2_OL = Oligodendrocyte, GABA = GABAergic Neuron), then if we determined based on spatial mapping that a cell type was localized to a specific subdivision of the brain or neuroanatomical region, we added that to the cell type classification. Many neuronal types did not map to specific spatial regions, so we only annotated them by cluster number and neurotransmitter. A dotplot of the strongest marker genes found by Seurat's "FindAllMarkers" function with only.pos = TRUE specified is shown in

Figure 2B.

Cell Type Composition Analysis

We calculated differences in cell population for each cluster using a logistic regression fitting a matrix of successes (number of cells in a cluster) versus failures (number of cells not in that cluster) by phase.

DEG Analysis

To calculate differentially expressed genes (DEGs), we used a similar approach to Parker et al.(5), Johnson et al. (60), and Connolly et al. (61). For each cluster, we used a negative binomial mixed regression using the "glmer" package (62). The response variable was the raw counts of the gene, the predictor variable was phase, and a random effect of individual ID was included. Dispersions were estimated using a gamma-Poisson model. We also included an offset term of the natural logarithm of the library sizes per nucleus to normalize variation in sequencing depth across the nuclei. For each gene, once the model was fit, type-III tests were computed and the false discovery rate (*q*-value) (55) for the "Phase" term was calculated within each cluster. A gene was considered a DEG if the *q*-value of type-III tests was < 0.05 and the model successfully converged.

Differentially Accessible Region (DAR) Analysis

Differentially accessible regions (DARs) were identified independently within each cluster using a pseudobulk approach implemented in edgeR. ATAC fragment counts were summed across nuclei from each individual to generate one pseudobulk accessibility profile per individual. Peaks detected in fewer than 10 individuals or with fewer than 80 total counts were excluded from analysis. Library sizes were normalized using the trimmed mean of M-values (TMM) method, dispersions were estimated with estimateDisp(), and quasi-likelihood generalized linear models were fit using glmQLFit(). Pairwise contrasts (M vs. F, M vs. D, and F vs. D) were tested using quasi-likelihood F-tests (glmQLFTest()), and peaks with a false discovery rate (FDR) < 0.20 in any comparison were considered differentially accessible.

Interpretation of Temporal Expression and Accessibility Patterns

Because sex change represents a temporal progression from males through intermediates to females, both DEGs and DARs were classified according to their pairwise differences among these three groups (Appendix SX). Features that differed between males and intermediates but not females were classified as "Early", whereas those differing between intermediates and females but not males were classified as "Late". Features with intermediate values between males and females were classified as "Progressive", whereas those uniquely elevated or reduced

in intermediates were classified as “Transient”. Directionality (Upregulated or Downregulated) was assigned according to whether expression or accessibility increased or decreased during sex change.

Integration of Differential Gene Expression and Chromatin Accessibility

To evaluate the relationship between differential gene expression and chromatin accessibility, we performed two complementary pseudobulk correlation analyses. In both analyses, RNA and ATAC counts were summed across nuclei from each individual within each cluster to generate pseudobulk profiles. Individuals represented by fewer than five nuclei within a cluster were excluded. RNA and ATAC libraries were normalized independently to log counts-per-million (logCPM), and Pearson and Spearman correlations were calculated across individuals. Correlation analyses were considered valid only when at least 18 individuals were available, RNA and ATAC counts were nonzero in at least 18 individuals, and both datasets exhibited non-zero variance.

For the DEG-centered analysis, each DEG was paired with all ATAC peaks overlapping the annotated gene body or a proximal promoter region (2 kb upstream and 500 bp downstream of the transcription start site). The peak exhibiting the strongest absolute Pearson correlation with gene expression was retained for each DEG. For the complementary DAR-centered analysis, each DAR was paired with its nearest annotated gene, and chromatin accessibility was correlated with expression of that gene across individuals. Pearson P-values from both analyses were corrected for multiple testing using the Benjamini-Hochberg false discovery rate procedure.

Machine Learning Analysis of Temporal Course of Sex Change

To estimate the temporal order of sex change in each of our multiome clusters, we used a machine learning approach to ask whether intermediates in a given cluster were more similar to males or females. For each cluster, we constructed a feature matrix in which rows corresponded to individuals and columns corresponded to DEGs or DARs that significantly differed between males and females. Then, using the Python package “scikit-learn”(63) we standardized each feature matrix (StandardScaler), encoded sex as a binary variable (male = 0, female = 1), and trained logistic regression classifiers (LogisticRegression; C = 10, solver = “lbfgs”). We then predicted the sex of each intermediate individual using predict_proba.

To restrict our analysis to high-confidence clusters, we additionally performed a leave-one-out procedure in which we iteratively refit the classifier while holding out one male or one female individual at a time and predicted its sex. We retained a cluster only if the mean predicted probability across held-out males was within 0.1 of 0 and the mean predicted probability across held-out females was within 0.1 of 1. We interpreted each cluster's mean prediction for intermediates as proxy for progression toward the female state at 6 months.

Subclustering 1_RGC

We subclustered 1_RGC using Seurat’s FindSubCluster function on the harmony weighted shared nearest neighbors graph with resolution 0.2 specified.

Cellular Maturity Inference

To infer relative cellular maturity, we used the R package “CytoTRACE”(64) to score each cell between 0 (most mature) and 1 (most immature) based on the diversity of its normalized RNA expression matrix. In order to assess cellular maturity differences in Figure 3D, we then took the mean CytoTRACE score for each individual in each subcluster and used a fixed effects linear model to fit CytoTRACE score by subcluster. Similarly, for Figure 4E, we fit individual means with phase as a fixed effect.

In analyses where we test if certain cell types differ by phase in cellular maturity, we instead used a linear mixed effects model fitting cell-level CytoTRACE scores with a fixed effect of nCount_RNA (the number of transcripts detected in the cell) detected in the cell and phase, gene(s) presence, and phase x gene presence as fixed effects along with individual as a random effect. This is necessary because CytoTRACE measures cellular maturity via the diversity of the cell’s transcriptome, however this is confounded when selecting a specific gene or set of genes for analysis. Including for the number of transcripts detected in each cell corrects for this

GO Analysis

Gene ontology (GO) analysis was performed using the “enricher” function from the R package “clusterProfiler” with “pAdjustMethod = “fdr” and p and q value cutoffs of 0.05. Because the *A. ocellaris* genome is poorly annotated, we used GO IDs and GO terms from the *Homo sapiens* genome. To do this we first mapped, human orthologs using Ensembl's precomputed homology annotations, retrieved via the R package “biomaRt” (aocellaris_gene_ensembl dataset). Since gene symbols and Ensembl homology calls are stored under separate BioMart attribute pages, we joined these using shared Ensembl transcript IDs. For a small number of genes with well-established nomenclature discrepancies between *A. ocellaris* and human annotations (e.g., kisspeptin, oxytocin/vasopressin, and their associated receptors), we manually corrected the assigned human ortholog based on gene identity rather than Ensembl's automated name matching.

GO Gene Expression Modules

GO gene expression modules were defined as all genes sharing the same GO ID as described above. To calculate a GO module score, we next used the R package “AUCell”(65) to first build gene expression rankings from the normalized RNA expression matrix (AUCell_buildRankings) and then calculate AUC scores for each cell (AUCell_calcAUC) given the set of genes included in the GO module described above. Differences in phase were calculated by taking the mean AUCell score for each individual and fitting the mean AUCell score by phase in a fixed effects linear model.

Data Availability

(...) I will make a Github repo when we are done

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